Traceability Matrix – Functional Requirements

User Story ID	Requirement ID	Requirements Description
US#10	FR1	The system shall allow a blood seeker to create a blood request.
US#10	FR2	The system shall send the blood request to the selected donor.
US#10	FR3	The system shall store and manage request information (request ID, seeker ID, donor ID, status, timestamp.
US#12	FR4	The system shall allow donors to accept or decline blood requests.
US#10	FR5	The system shall update the request status (pending, accepted, declined).
US#11	FR6	The system shall notify donors when they receive a blood request.
US#13	FR7	The system shall notify blood seekers when a donor accepts or declines a request.
US#11	FR8	The system shall manage system notifications (e.g., admin alerts, updates).
US#14	FR9	The system shall enable communication between donors and seekers once a request is accepted.
US#14	FR10	The system shall allow sending and receiving messages in real time.
US#14	FR11	The system shall store and display message history for both users.
US#17	FR12	The system shall allow administrators to view and manage registered users.
US#18	FR13	The system shall allow administrators to manage hospitals and donation centers.
US#19	FR14	The system shall allow administrators to review reported accounts and take appropriate actions.
US#20	FR15	The system shall allow administrators to monitor and manage blood requests and platform activities.

Traceability Matrix – Non-Functional Requirements

Requirement ID	Category	Requirements Description
NFR1	Performance	Notifications shall be delivered within 5 seconds of a request being sent.
NFR2	Performance	The messaging system shall support at least 100 concurrent conversations without delay.
NFR3	Security	All messages shall be encrypted to protect user privacy.
NFR4	Security	Only authenticated users shall be able to send or receive blood requests.
NFR5	Security	Admin dashboard access shall be restricted to authorized administrators only.
NFR6	Usability	The request creation form shall be simple and intuitive, requiring no more than 3 steps.
NFR7	Usability	Notifications shall be displayed clearly in the user interface with distinct icons.
NFR8	Usability	The admin dashboard shall provide a clean interface with easy navigation menus.
NFR9	Reliability	The system shall maintain 99% uptime during active hours.
NFR10	Scalability	The system shall scale to support up to 10,000 registered users.